CHESS COMPANION

TEAM MEMBERS



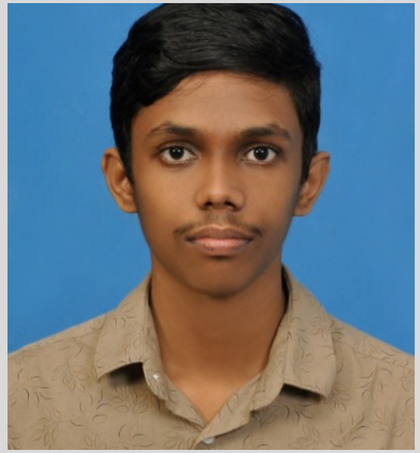
Thejaka Seneviratne
244195V



Sithum Sandaruwan
244189G



Avindu Thilakarathne
244206J



Sahan Wijerathne
244227A



Harishan Nirmalan
244074B

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Introduction

Autonomous Chessboard is a smart physical chessboard that combines sensors, motors, electronics, and AI. In a game, a human plays one side while the Stockfish chess engine plays the other. Hall-effect sensors detect the positions of the pieces, while an ESP32-S3 controls a moving system and electromagnet to move the engine's pieces automatically. The board has 256 LEDs that show legal moves, captures, special moves, warnings, and game results. Speakers provide voice feedback and coaching through a Gemini mentor, which can analyze positions, explain mistakes, change the difficulty level, and teach chess openings. A web application connects and controls all these features. Offline engine support and fully standalone play are also being developed.

AIM

The aim of this project is to develop an autonomous physical chessboard that allows a human player to play against an AI opponent in a realistic and interactive way. By combining a real chessboard with artificial intelligence, the project aims to make chess practice more engaging, reduce dependence on screens, and help players improve their skills.

OBJECTIVES

- To build a physical chessboard that can move pieces automatically using a hidden gantry and electromagnet.
- To use Hall-effect sensors to detect the position of every chess piece.
- To integrate Stockfish so the system can play against a human at different difficulty levels.
- To use LEDs to show legal moves, captures, warnings, and game results.
- To develop a web application for game settings and live game information.
- To add speakers and a Gemini voice mentor for chess explanations and coaching.
- To combine all hardware and software components into one reliable chess-playing system.

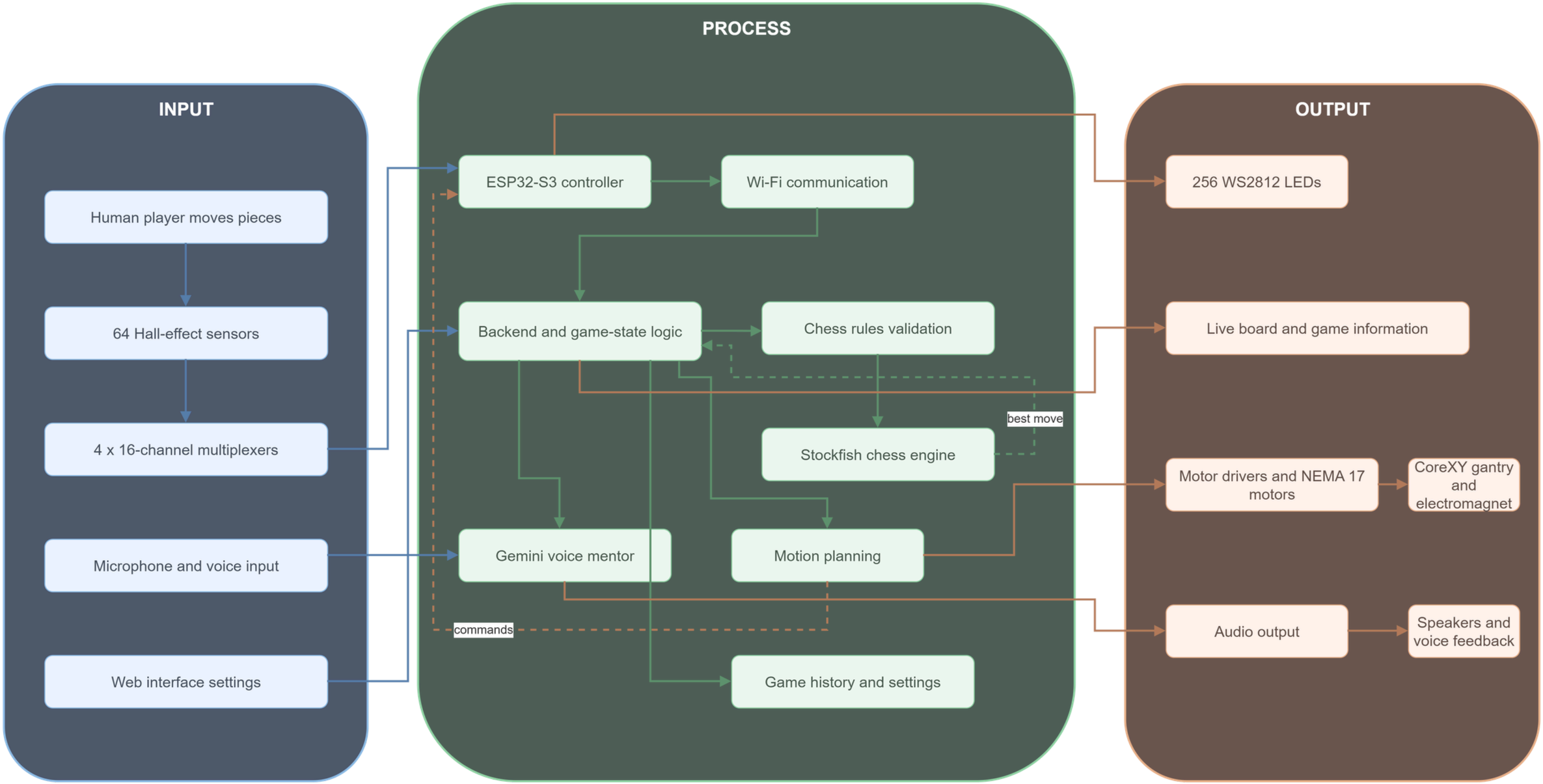
PROBLEM

- Finding a human opponent is not always easy, making solo chess practice difficult.
- Digital chess applications require long periods of screen use, which can cause eye strain.
- Digital platforms do not provide the physical experience of moving real chess pieces.
- Traditional chessboards usually do not provide an automatic opponent, move guidance, or learning support.
- There is a need for a system that combines the physical experience of chess with the benefits of AI.

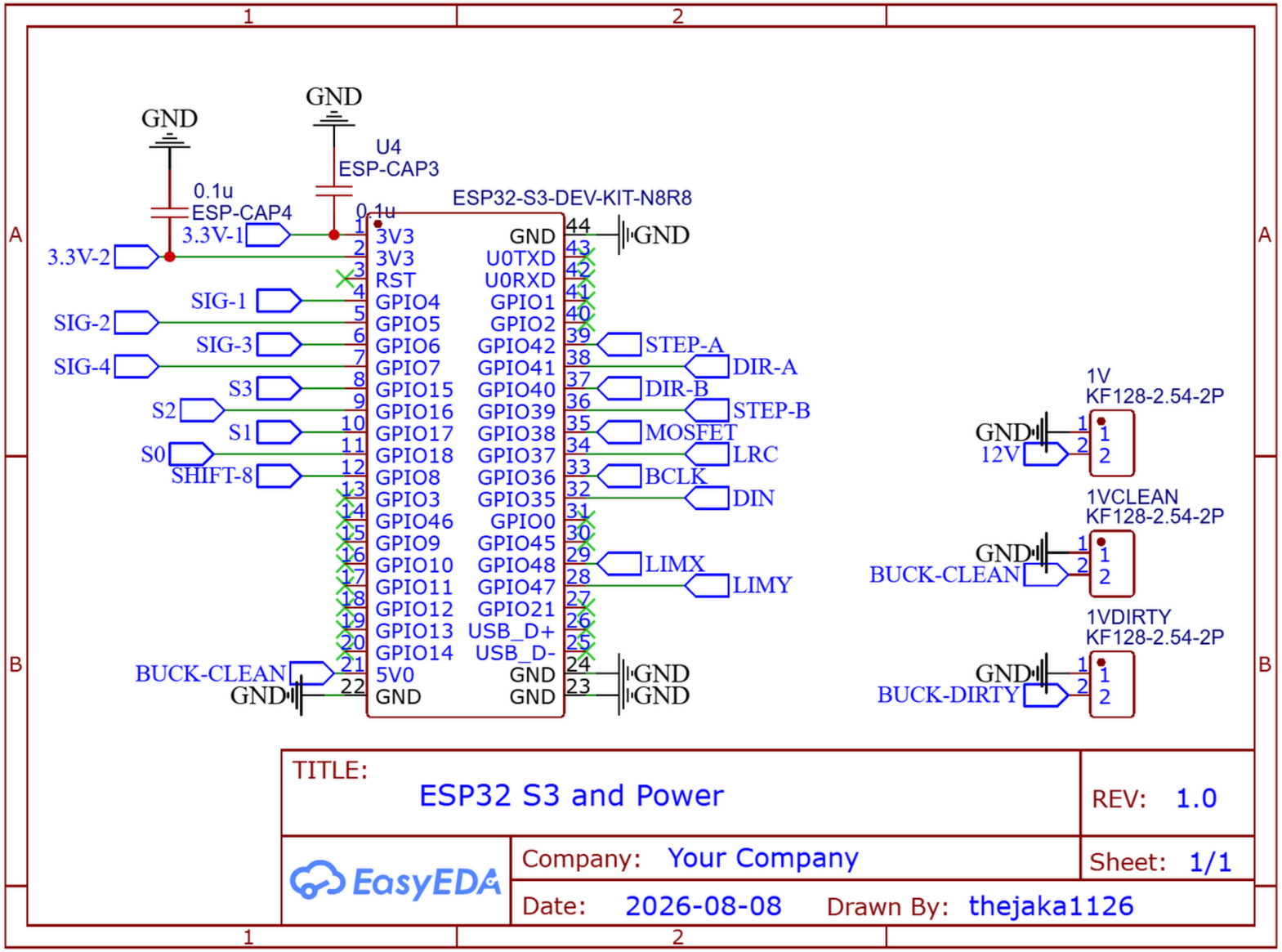
SOLUTION

- Provide an AI opponent using Stockfish so players can practise chess without needing another person.
- Use a physical board with real chess pieces to create a more hands-on experience.
- Use sensors to detect player moves and automatically move the AI’s pieces.
- Use LEDs to show legal moves, captures, warnings, and other game information.
- Use speakers and a Gemini voice mentor to explain moves and help players learn.
- Use a web application to manage the game, settings, difficulty level, and live board state.
- Reduce the need for continuous screen use while keeping the benefits of a digital chess system.

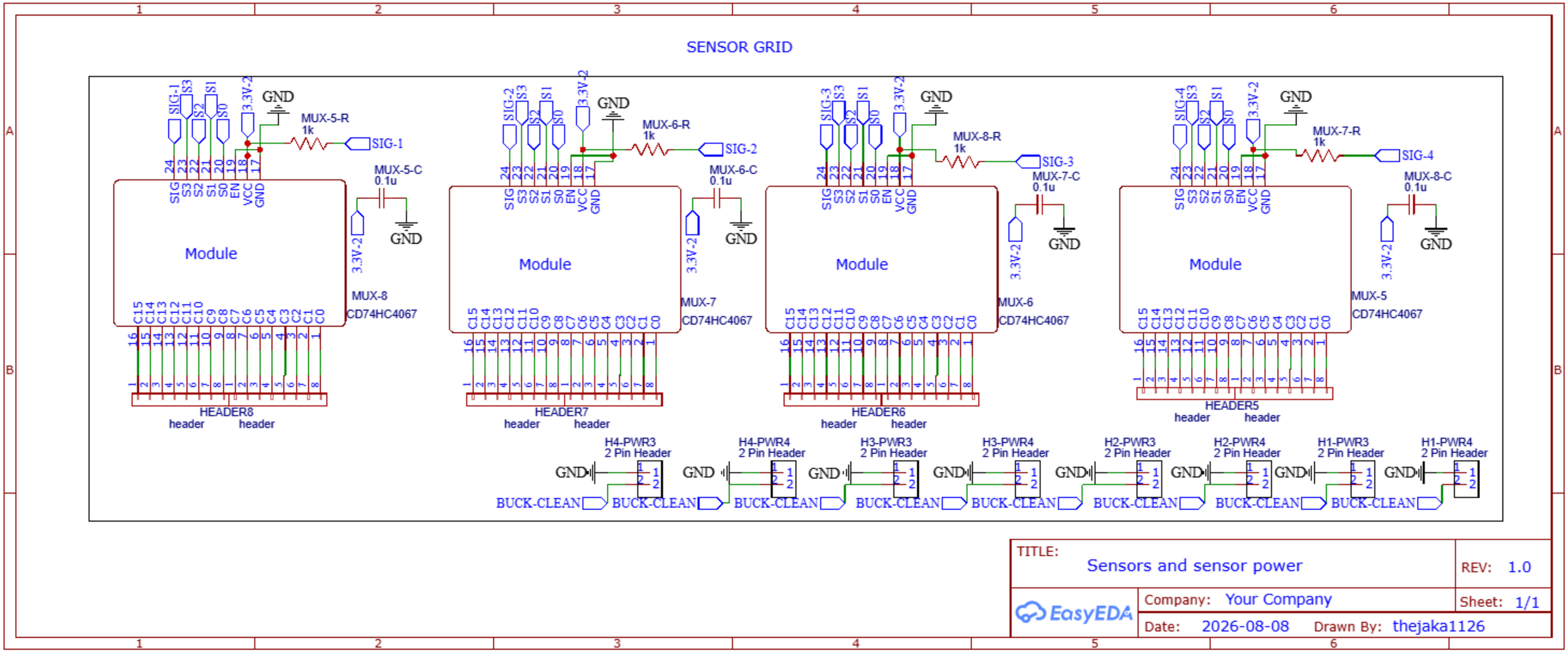
BLOCK DIAGRAM



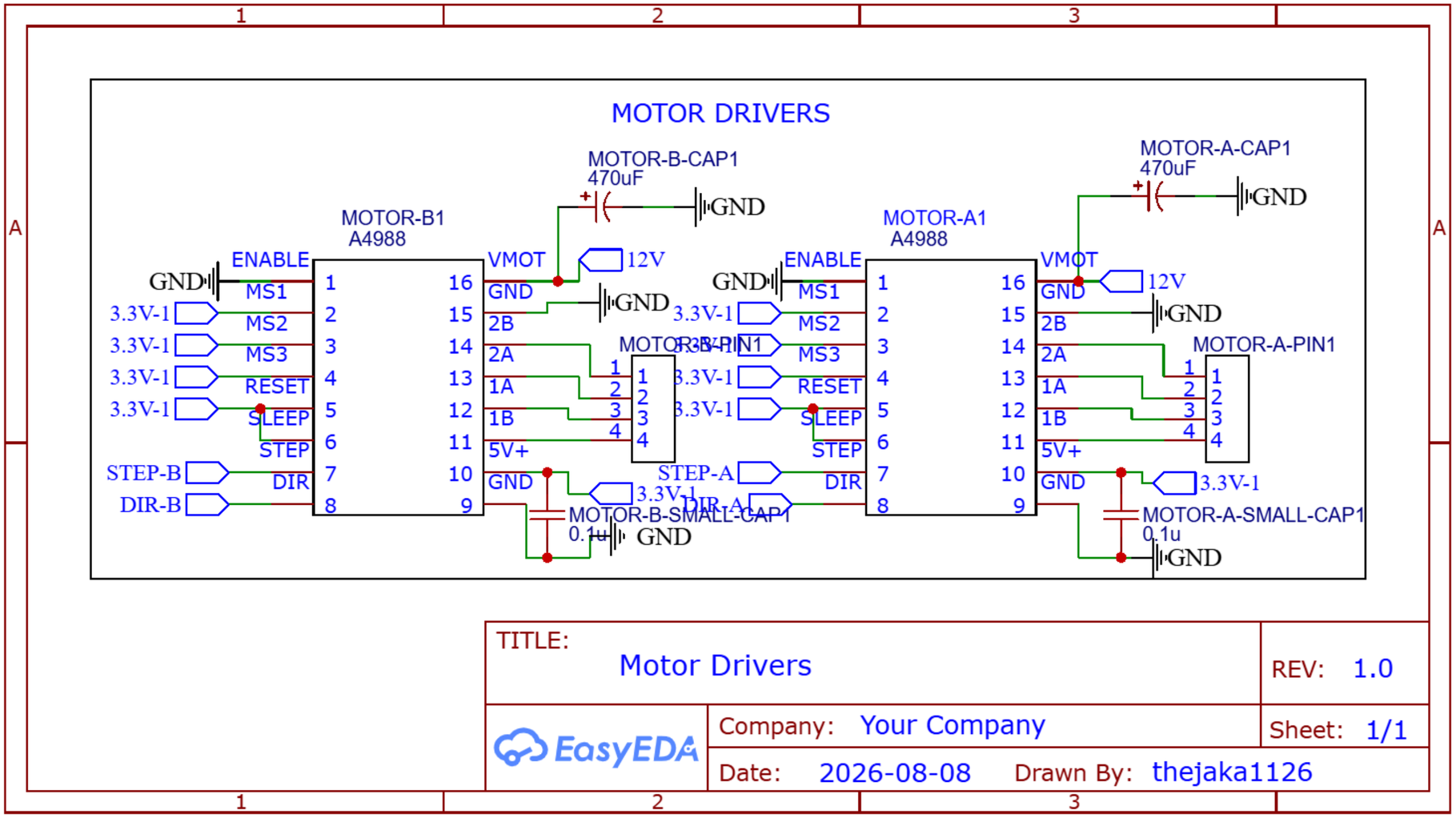
SCHEMATICS


GND
GND
U4
ESP-CAP3
0.1u
ESP-CAP4
0.1u
ESP32-S3-DEV-KIT-N8R8
3.3V-1
3.3V-2
SIG-1
SIG-2
SIG-3
SIG-4
S3
S2
S1
S0
SHIFT-8
BUCK-CLEAN
GND
3V3
3V3
RST
GPIO4
GPIO5
GPIO6
GPIO7
GPIO15
GPIO16
GPIO17
GPIO18
GPIO8
GPIO3
GPIO46
GPIO9
GPIO10
GPIO11
GPIO12
GPIO13
GPIO14
5V0
GND
GND
U0TXD
U0RXD
GPIO1
GPIO2
GPIO42
GPIO41
GPIO40
GPIO39
GPIO38
GPIO37
GPIO36
GPIO35
GPIO0
GPIO45
GPIO48
GPIO47
GPIO21
USB_D+
USB_D-
GND
GND
STEP-A
DIR-A
DIR-B
STEP-B
MOSFET
LRC
BCLK
DIN
LIMX
LIMY
1V
KF128-2.54-2P
GND
12V
1VCLEAN
KF128-2.54-2P
GND
BUCK-CLEAN
1VDIRTY
KF128-2.54-2P
GND
BUCK-DIRTY
TITLE:
ESP32 S3 and Power
REV: 1.0
EasyEDA
Company: Your Company
Sheet: 1/1
Date: 2026-08-08
Drawn By: thejaka1126

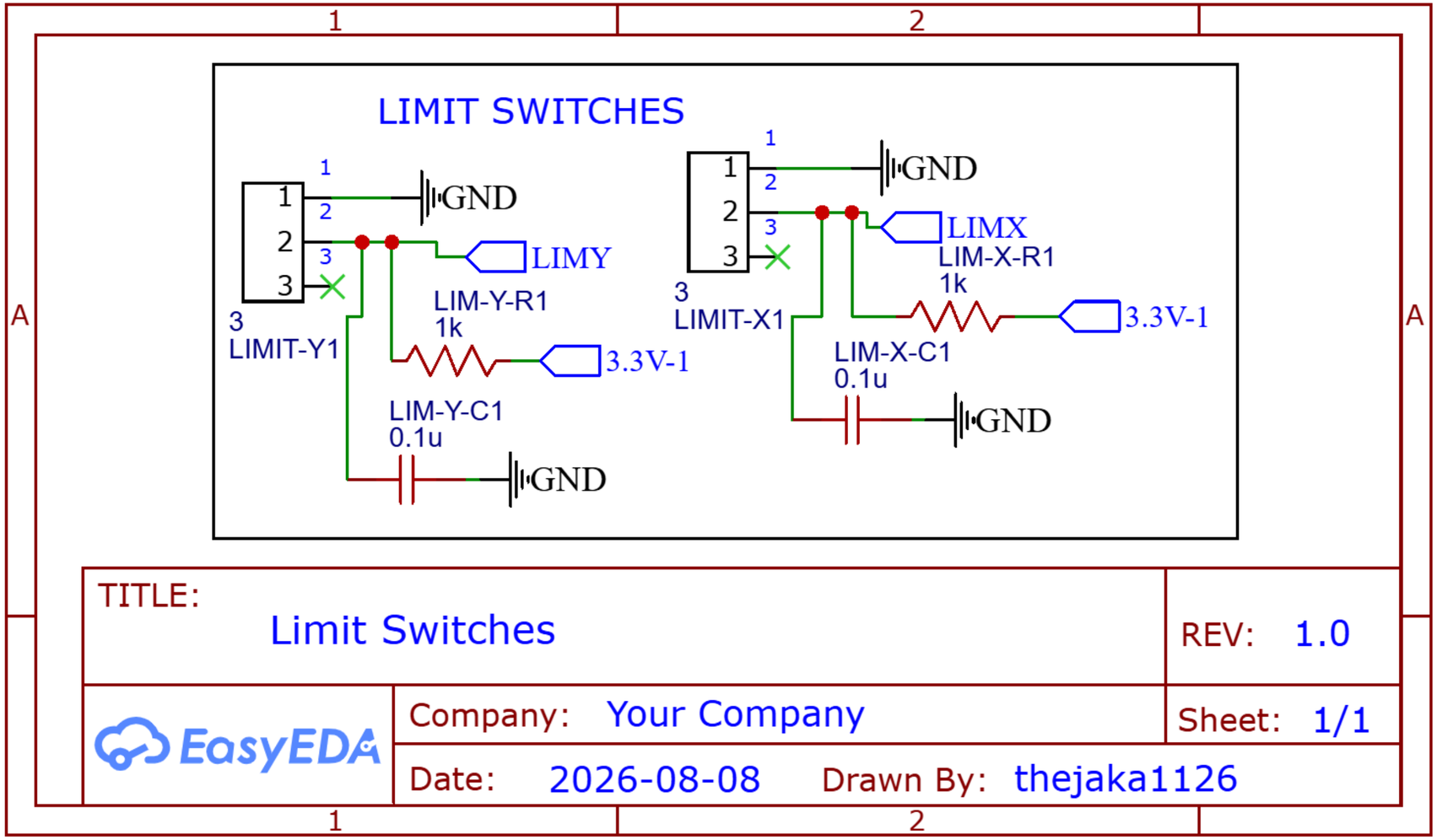
SCHEMATICS


SENSOR GRID
MUX-5-R
1k
SIG-1
MUX-5-C
0.1u
Module
MUX-8
CD74HC4067
HEADER8
MUX-6-R
1k
SIG-2
MUX-6-C
0.1u
MUX-7
CD74HC4067
HEADER7
MUX-8-R
1k
SIG-3
MUX-7-C
0.1u
MUX-6
CD74HC4067
HEADER6
MUX-7-R
1k
SIG-4
MUX-8-C
0.1u
MUX-5
CD74HC4067
HEADER5
GND
3.3V-2
H4-PWR3
H4-PWR4
H3-PWR3
H3-PWR4
H2-PWR3
H2-PWR4
H1-PWR3
H1-PWR4
2 Pin Header
BUCK-CLEAN
TITLE:
Sensors and sensor power
REV: 1.0
Company: Your Company
Sheet: 1/1
Date: 2026-08-08
Drawn By: thejaka1126
EasyEDA

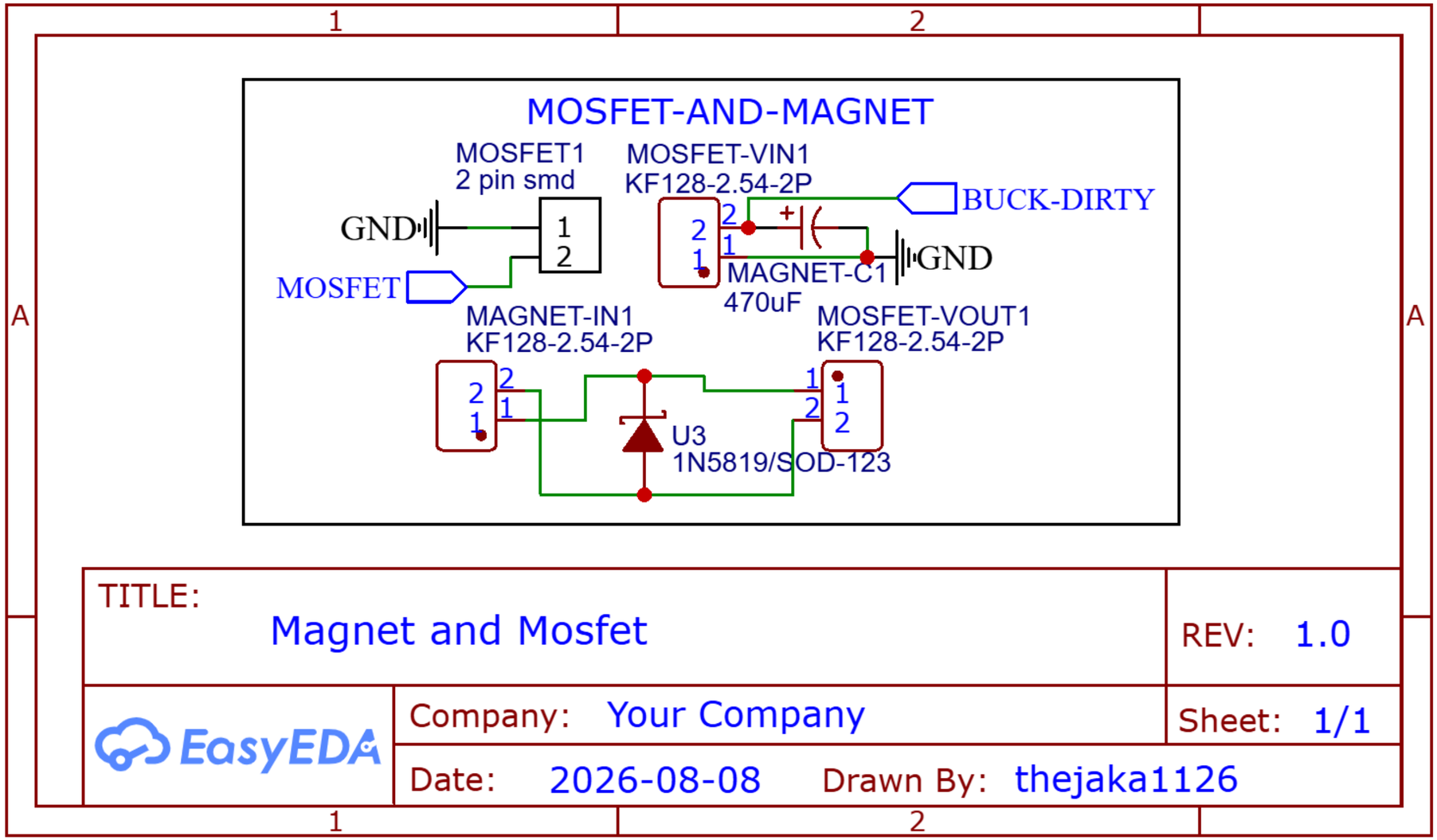
SCHEMATICS


MOTOR DRIVERS
MOTOR-B1
A4988
MOTOR-B-CAP1
470uF
MOTOR-A1
A4988
MOTOR-A-CAP1
470uF
ENABLE
MS1
MS2
MS3
RESET
SLEEP
STEP
DIR
VMOT
GND
2B
2A
1A
1B
5V+
12V
3.3V-1
STEP-B
DIR-B
STEP-A
DIR-A
MOTOR-B-PIN1
MOTOR-A-PIN1
MOTOR-B-SMALL-CAP1
MOTOR-A-SMALL-CAP1
0.1u
TITLE:
Motor Drivers
REV: 1.0
Company: Your Company
Sheet: 1/1
Date: 2026-08-08
Drawn By: thejaka1126
EasyEDA

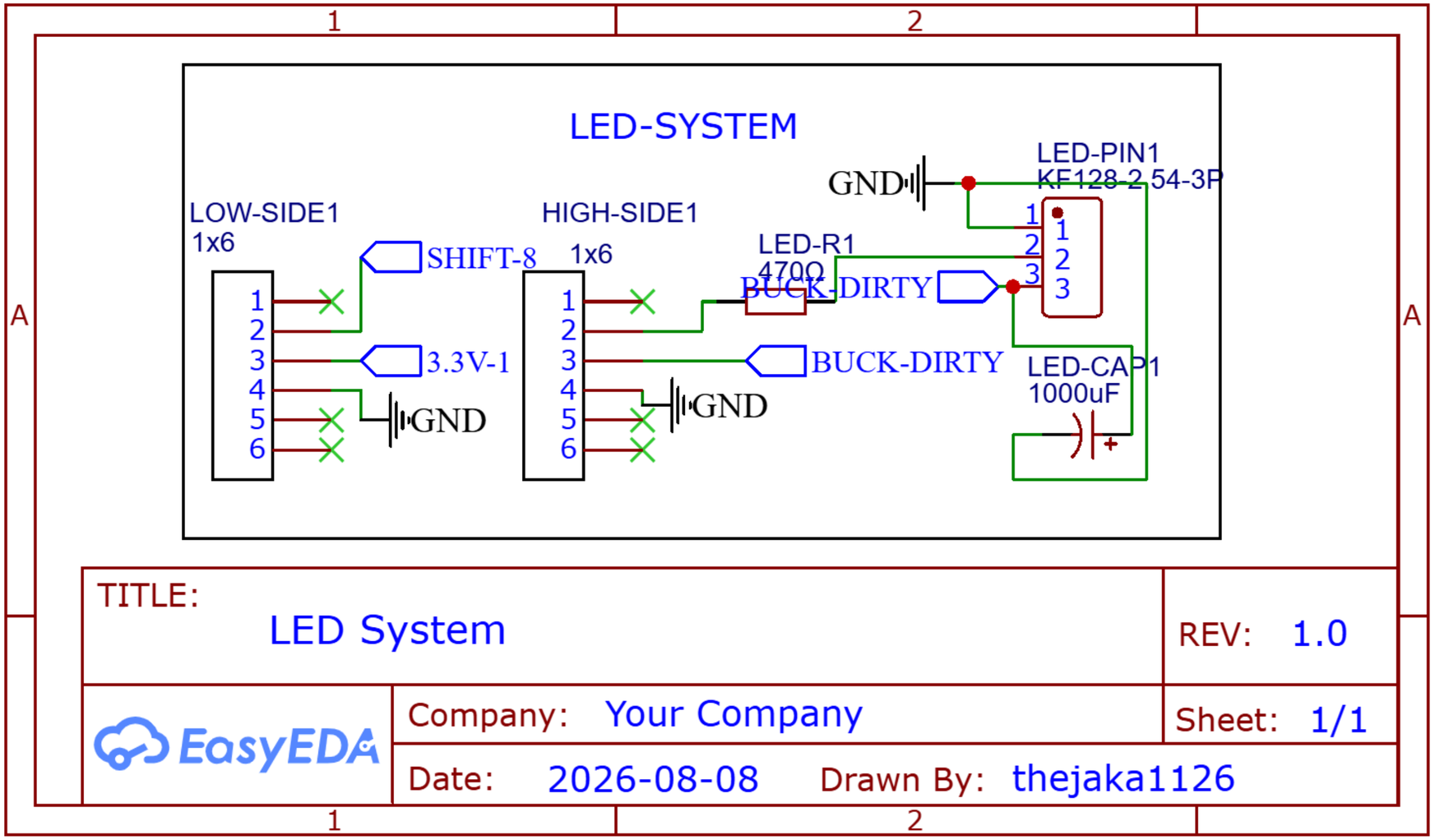
SCHEMATICS


LIMIT SWITCHES
GND
LIMY
LIM-Y-R1
1k
3.3V-1
LIM-Y-C1
0.1u
GND
3
LIMIT-Y1
GND
LIMX
LIM-X-R1
1k
3.3V-1
LIM-X-C1
0.1u
GND
3
LIMIT-X1
TITLE:
Limit Switches
REV: 1.0
EasyEDA
Company: Your Company
Sheet: 1/1
Date: 2026-08-08
Drawn By: thejaka1126

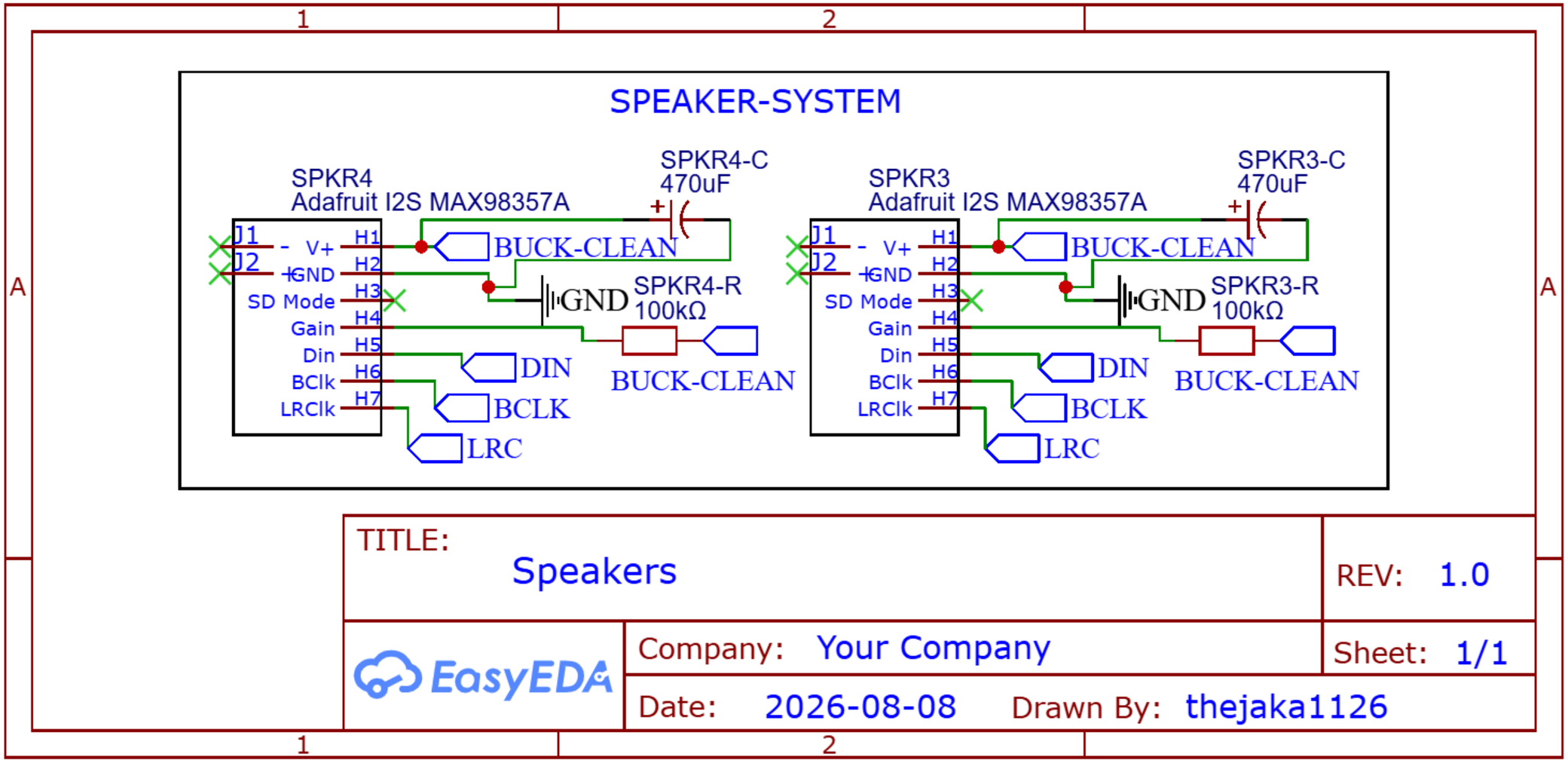
SCHEMATICS


MOSFET-AND-MAGNET
MOSFET1
2 pin smd
GND
MOSFET
MOSFET-VIN1
KF128-2.54-2P
BUCK-DIRTY
GND
MAGNET-C1
470uF
MAGNET-IN1
KF128-2.54-2P
MOSFET-VOUT1
KF128-2.54-2P
U3
1N5819/SOD-123
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Magnet and Mosfet
REV: 1.0
Company: Your Company
Sheet: 1/1
Date: 2026-08-08
Drawn By: thejaka1126
EasyEDA

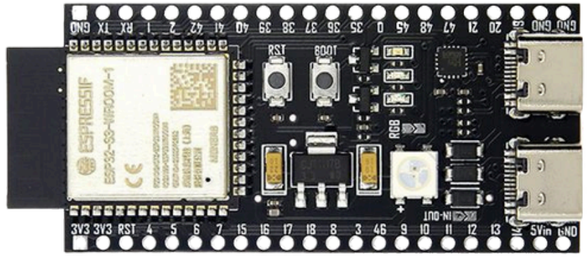
SCHEMATICS


LED-SYSTEM
LOW-SIDE1
1x6
SHIFT-8
3.3V-1
GND
HIGH-SIDE1
1x6
LED-R1
470Ω
BUCK-DIRTY
BUCK-DIRTY
GND
LED-PIN1
KF128-2.54-3P
GND
LED-CAP1
1000uF
TITLE:
LED System
REV: 1.0
EasyEDA
Company: Your Company
Sheet: 1/1
Date: 2026-08-08
Drawn By: thejaka1126

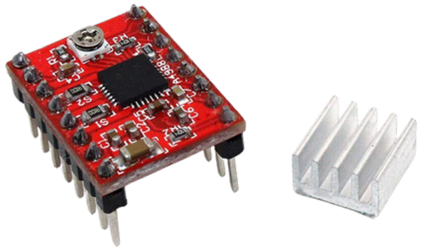
SCHEMATICS

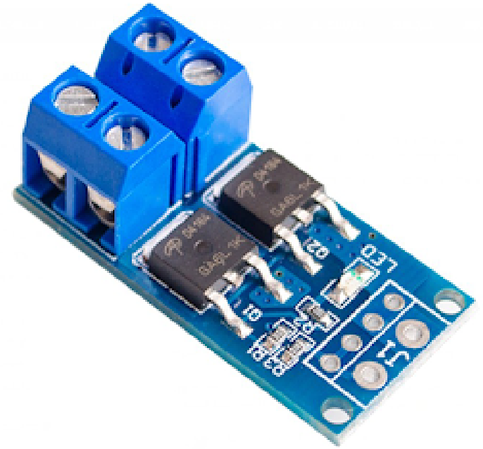

1
2
A
SPEAKER-SYSTEM
SPKR4
Adafruit I2S MAX98357A
SPKR4-C
470uF
J1
J2
- V+
GND
SD Mode
Gain
Din
BClk
LRClk
H1
H2
H3
H4
H5
H6
H7
BUCK-CLEAN
GND
SPKR4-R
100kΩ
DIN
BUCK-CLEAN
BCLK
LRC
SPKR3
Adafruit I2S MAX98357A
SPKR3-C
470uF
SPKR3-R
100kΩ
TITLE:
Speakers
REV: 1.0
EasyEDA
Company: Your Company
Sheet: 1/1
Date: 2026-08-08
Drawn By: thejaka1126

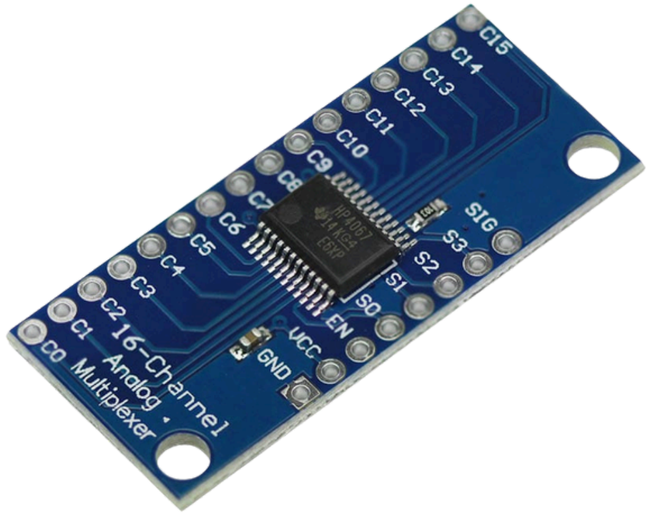
COMPONENTS

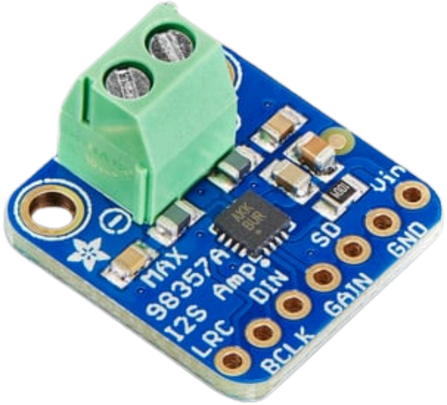

ESP32-S3-WROOM-1
RST
BOOT
RGB
3V3 3V3 RST
5Vin GND

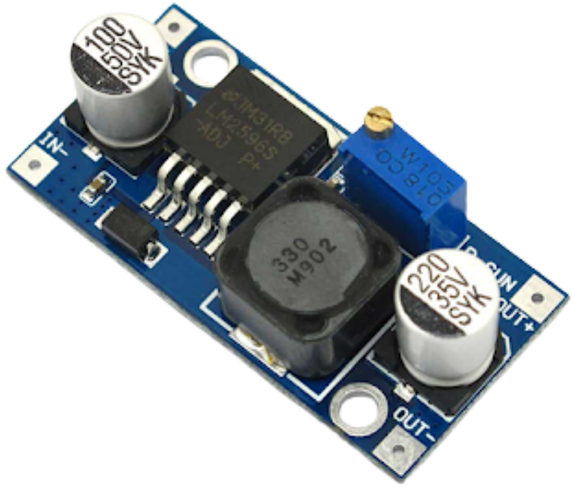


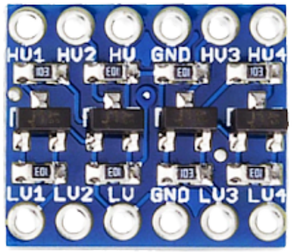





16-Channel
Analog
Multiplexer
SIG
S3
S2
S1
S0
EN
VCC
GND


MAX
98357A
I2S Amp
DIN
BCLK
LRC
GAIN
SD
Vin
GND


IN-
IN+
OUT-
OUT+
100
220

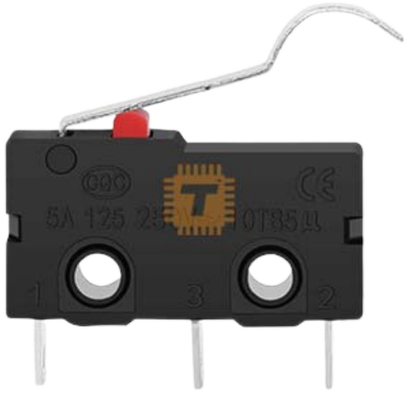

HV1 HV2 HV GND HV3 HV4
LV1 LV2 LV GND LV3 LV4

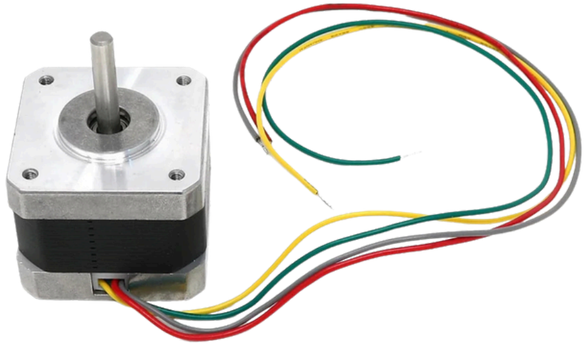
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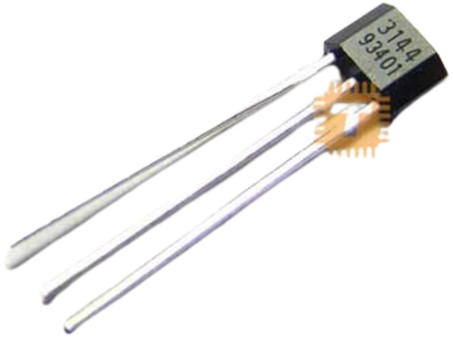




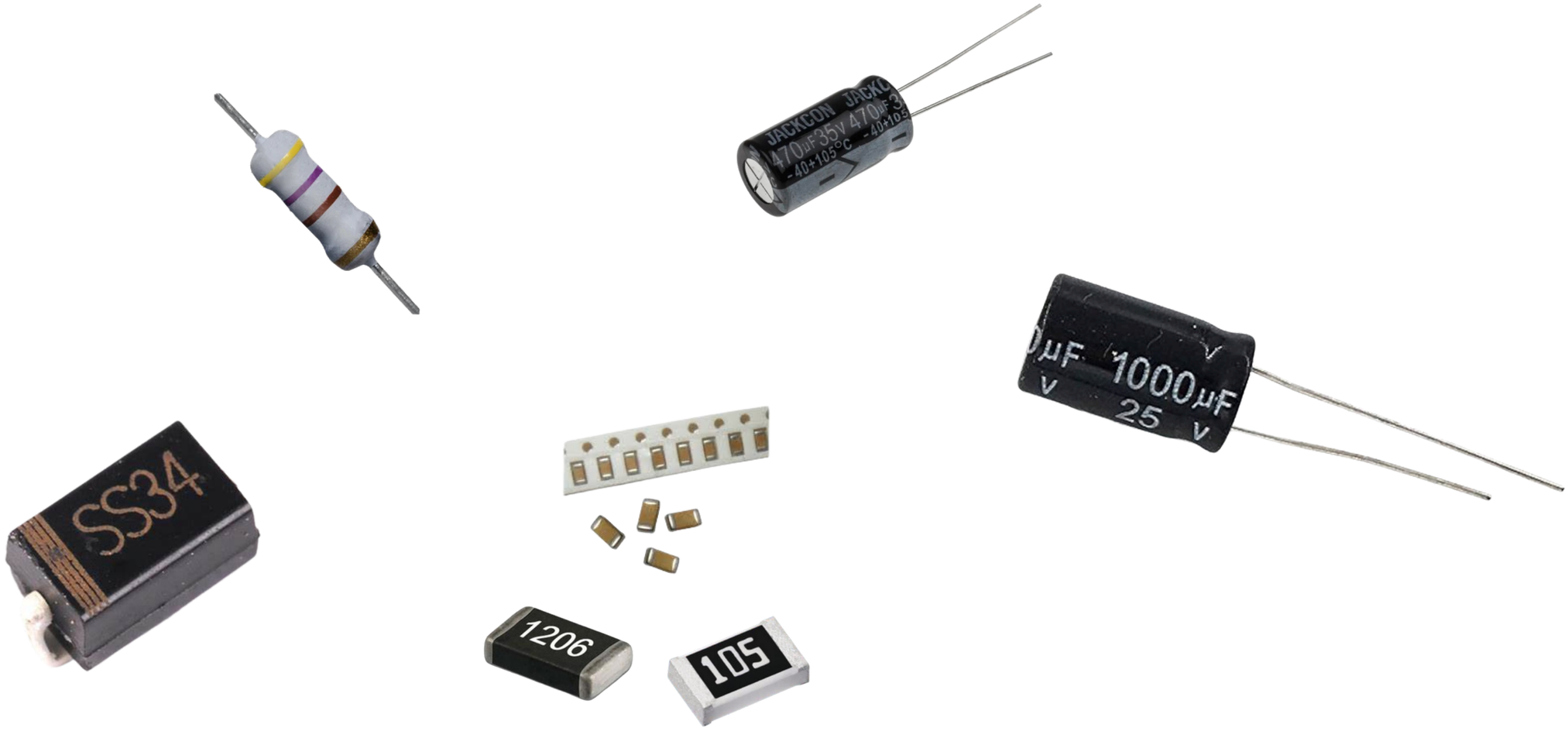








COMPONENTS



COMPONENTS



PRODUCT VIEW







WORK DISTRIBUTION

WORK BREAKDOWN STRUCTURE

The project was divided into 12 main sections. Each section was led by a team member and supported by other members where needed.

1. Enclosure — Sahan
2. Stepper Motors and Motor Control — Avindu
3. Electromagnet System — Harishan
4. CoreXY Gantry — Thejaka
5. Limit Switches — Sahan
6. Speaker System — Sithum
7. LED System — Sithum
8. Sensor System — Thejaka
9. PCB and Circuit Design — Avindu
10. Power Supply — Sahan
11. Chessboard Sticker and Grid — Thejaka
12. Basic Web Application — Thejaka

WBS Code	Task	Assigned member
1.1	Confirm box dimensions and internal layout	Sahan
1.2	Measure and mark MDF panels	Sahan
1.3	Cut panels and drill mounting holes	Sahan
1.4	Assemble the box and check alignment	Sahan
1.5	Sand the box and prepare cable openings	Sahan
1.6	Install internal mounts and check the fit	Avindu
1.7	Assist with mounting and cable clearance	Harishan
2.1	Identify stepper motors, drivers and wiring	Avindu
2.2	Test basic stepper motor movement	Avindu, Thejaka
2.3	Calculate steps for each chessboard square	Avindu
2.4	Control motors and map board positions	Avindu, Thejaka
2.5	Fix direction, alignment and missed-step issues	Avindu
2.6	Prepare and secure motor-driver wiring	Thejaka, Harishan

WBS Code	Task	Assigned member
3.1	Connect the MOSFET and electromagnet for testing	Harishan
3.2	Test magnet switching and movement	Harishan
3.3	Combine the electromagnet with the gantry	Harishan, Thejaka
3.4	Test pickup and drop of magnetic pieces	Harishan
3.5	Insulate and secure electromagnet wires	Sithum
3.6	Prepare magnet mounting hardware and wire strain relief	Harishan
4.1	Take dimensions and determine grid spacing	Thejaka
4.2	Design the CoreXY layout	Thejaka
4.3	Build the gantry frame	Thejaka
4.4	Install rods, bearings, belts and motor mounts	Thejaka, Sahan, Harishan
4.5	Align the frame and check smooth movement	Thejaka, Avindu
4.6	Measure rods and tighten mechanical fasteners	Sahan

WBS Code	Task	Assigned member
5.1	Decide the limit switch positions	Sahan
5.2	Mount the limit switches	Sahan
5.3	Connect and test the limit-switch wiring	Sithum
5.4	Test homing and adjust trigger positions	Sahan
5.5	Label and secure limit-switch wires	Thejaka
6.1	Check speaker operation and range	Sithum
6.2	Test stored sounds and audio	Sithum
6.3	Mount speakers and route the wires	Sithum
6.4	Check volume, sound quality and connectors	Sithum
7.1	Test WS2812 LED strips with basic code	Sithum
7.2	Design the final LED grid	Sithum
7.3	Solder the 64 LED grids and inspect the joints	Sithum, Sahan, Harishan
7.4	Connect LED power and data lines	Sithum, Harishan
7.5	Test the LED grids and repair faults	Sithum
7.6	Label LED order and verify square positions	Avindu

WBS Code	Task	Assigned member
8.1	Test 16 Hall sensors with one multiplexer	Thejaka
8.2	Design the sensor-line and veroboard layout	Thejaka
8.3	Solder the sensors and connect ribbon wires	Thejaka, Sithum
8.4	Connect the multiplexers and fix issues	Thejaka
8.5	Test the complete 64-square sensor system	Thejaka, Avindu
8.6	Label sensor rows, columns and wires	Sithum
9.1	Identify circuit components and connections	Avindu
9.2	Design the circuit schematic	Avindu
9.3	Plan wire routing and PCB tracks	Avindu, Thejaka
9.4	Adjust the design after testing	Avindu
9.5	Check connectors, labels and power-ground continuity	Sithum, Harishan

WBS Code	Task	Assigned member
10.1	Check datasheets and power requirements	Sahan
10.2	Separate the 12 V and 5 V power rails	Sahan
10.3	Design a compact voltage-management module	Sahan
10.4	Assemble the power-management and distribution system	Sahan, Sithum
10.5	Measure voltages under load and fix issues	Sahan, Sithum
10.6	Cut, strip and label power wires	Sahan, Sithum
11.1	Check chessboard size standards	Thejaka
11.2	Select the square size and determine positions	Thejaka
11.3	Design and print the chessboard grid	Thejaka, Avindu
11.4	Apply and align the chessboard sticker	Sahan
11.5	Check sticker alignment with the LEDs and sensors	Thejaka
12.1	Create a simple board-status interface	Thejaka
12.2	Connect the app to the ESP32 and display moves	Thejaka
12.3	Add basic game-state and engine communication	Thejaka
12.4	Perform basic app testing and documentation	Thejaka, Sithum
12.5	Test the app with the physical chessboard	Thejaka

TEAM ALLOCATION

Thejaka Seneviratne 244195V—

- Lead: 4.1–4.5, 8.1–8.5, 11.1–11.3, 11.5, 12.1–12.5
- Support: 2.2, 2.4, 2.6, 3.3, 5.5, 9.3

Avindu Thilakarathne 244206J—

- Lead: 2.1–2.5, 9.1–9.4
- Support: 1.6, 4.5, 7.6, 8.5, 11.3

Harishan Nirmalan 244074B —

- Lead: 3.1–3.4, 3.6
- Support: 1.7, 2.6, 4.4, 7.3, 7.4, 9.5

Sahan Wijeratne 244227A —

- Lead: 1.1–1.5, 5.1–5.2, 5.4, 10.1–10.6
- Support: 4.4, 4.6, 7.3, 11.4

Sithum Sandaruwan 244189G —

- Lead: 6.1–6.4, 7.1–7.5, 9.5
- Support: 3.5, 5.3, 8.3, 8.6, 10.4–10.6, 12.4